

SAMPLE QUESTIONS

Q1) A precision engineering firm wants to minimize the surface roughness (Ra) of aluminum

workpieces produced by a CNC milling machine. The influencing factors considered are cutting speed (A), feed rate (B), and depth of cut (C), each at three levels.

Explain how the Taguchi design approach can be applied to this optimization problem.

Choose a suitable orthogonal array (L9), identify the factor levels, and show how to calculate

the S/N ratio for the “smaller-the-better” quality characteristic.

Discuss how the Taguchi analysis assists in identifying the optimal parameter settings that

minimize roughness while ensuring process consistency under varying conditions.

Provide a numerical example and interpret the results to demonstrate improvement in process

robustness and performance.

PROBLEM RESTATEMENT (TAGUCHI VIEW)

Goal: **minimize surface roughness (Ra)** of aluminium milling parts.

Factors (3 each, three levels):

- A — Cutting speed (m/min): 100, 150, 200
- B — Feed rate (mm/rev): 0.10, 0.20, 0.30
- C — Depth of cut (mm): 0.5, 1.0, 1.5

Choose an orthogonal array: **L9 (3³)** (suitable for 3 factors × 3 levels).

2) L9 orthogonal array (layout used)

Standard L9 combination (levels 1–3 for A,B,C):

Run A B C

1	1	1	1
2	1	2	2
3	1	3	3
4	2	1	2
5	2	2	3
6	2	3	1
7	3	1	3
8	3	2	1
9	3	3	2

(These are the standard orthogonal combinations used in Taguchi L9.)

3) Quality characteristic & S/N ratio formula

For “smaller-the-better” (Ra), Taguchi S/N ratio is:

$$S/N_{STB} = -10 \log_{10} \left(\frac{1}{n} \sum_{i=1}^n y_i^2 \right)$$

where y_i are the replicate measurements for a run and n is the number of replicates.

Larger S/N (less negative) → better (lower mean and lower variability) for smaller-the-better.

4) Numerical example (simulated measurements & calculations)

I generated 3 replicates per run (so 9 runs × 3 replicates = 27 measurements). The measured Ra (three replicates per run), mean Ra and S/N (smaller-the-better) for each run are:

Run	A (levels)	B (levels)	C (levels)	Speed (m/min)	Feed (mm/rev)	Depth (mm)	Replicates (Ra)	Mean Ra	S/N (dB)
1	A1	B1	C1	100	0.10	0.5	0.6799, 0.6823	0.6672, 0.6767	3.3915
2	A1	B2	C2	100	0.20	1.0	0.8505, 0.8150	0.8153, 0.8270	1.6478
3	A1	B3	C3	100	0.30	1.5	1.0016, 0.9606	0.9853, 0.9825	0.1519
4	A2	B1	C2	150	0.10	1.0	0.6809, 0.6606	0.6607, 0.6674	3.5111
5	A2	B2	C3	150	0.20	1.5	0.8248, 0.7855	0.7817, 0.7974	1.9643
6	A2	B3	C1	150	0.30	0.5	0.8388, 0.8562	0.8297, 0.8416	1.4972
7	A3	B1	C3	200	0.10	1.5	0.6518, 0.6996	0.6418, 0.6643	3.5465
8	A3	B2	C1	200	0.20	0.5	0.6955, 0.6712	0.7014, 0.6894	3.2285
9	A3	B3	C2	200	0.30	1.0	0.8391, 0.8266	0.8522, 0.8394	1.5196

(Values were generated with small experimental noise to simulate real data. I kept the math exact for S/N using the formula above.)

Overall (grand) means across the 9 runs:

- Grand mean S/N = **2.2732 dB**
- Grand mean Ra = **0.77620 μm**

5) Taguchi main effects (S/N means by level) — decision table

Compute average S/N (and mean Ra) at each level for each factor:

S/N (mean) by factor level

A (Cutting speed)

- A1 (100) : S/N = **1.7304** ; mean Ra = 0.8288
- A2 (150) : S/N = **2.3242** ; mean Ra = 0.7688
- A3 (200) : S/N = **2.7648** ; mean Ra = 0.7311

B (Feed)

- B1 (0.10) : S/N = **3.4830** ; mean Ra = 0.6695
- B2 (0.20) : S/N = **2.2802** ; mean Ra = 0.7713
- B3 (0.30) : S/N = **1.0562** ; mean Ra = 0.8878

C (Depth)

- C1 (0.50) : S/N = **2.7057** ; mean Ra = 0.7359
- C2 (1.00) : S/N = **2.2262** ; mean Ra = 0.7780
- C3 (1.50) : S/N = **1.8876** ; mean Ra = 0.8147

Selection rule (Taguchi): choose the level for each factor that gives the **largest S/N** (since larger S/N means better robustness for the smaller-the-better case).

Thus:

- Best A level = **A3** (200 m/min)
- Best B level = **B1** (0.10 mm/rev)
- Best C level = **C1** (0.50 mm)

Recommended optimal combination: A3 B1 C1 $\rightarrow$ (200 m/min, 0.10 mm/rev, 0.5 mm)

6) Predicted performance (Taguchi confirmation calculation)

Taguchi predicts the S/N (and the mean response) of the optimal combination by adding main effects to the grand mean.

Predicted S/N for A3B1C1:

$$\text{Pred_S/N} = \text{Grand_S/N} + (\bar{S/N}_{A3} - \text{Grand_S/N}) + (\bar{S/N}_{B1} - \text{Grand_S/N}) + (\bar{S/N}_{C1} - \text{Grand_S/N})$$

Using the numbers above:

- Grand S/N = 2.2732 dB
- A3 = 2.7648 ; B1 = 3.4830 ; C1 = 2.7057

Predicted S/N $\approx$ **4.4073 dB**

Predicted mean Ra (same additive approach using mean Ra by level):

- Grand mean Ra = 0.77620 μm
- A3 Ra = 0.7311 ; B1 Ra = 0.6695 ; C1 Ra = 0.7359

Predicted Ra $\approx$ **0.58405 μm**

Improvement vs current (grand mean):

- Absolute reduction in Ra $\approx 0.77620 - 0.58405 =$ **0.19215 μm**
- Percentage reduction $\approx (0.19215 / 0.77620) \times 100 \approx$ **24.8% reduction in mean Ra**

So Taguchi predicts roughly a **~25% improvement** in mean Ra and a higher S/N (more robust/lower variance) at A3B1C1.

7) Which factors matter most? (ranking & contribution)

A simple Taguchi ranking via **range (max S/N – min S/N)** shows relative effects:

- **Feed (B):** largest effect (range ≈ 2.427 dB) $\rightarrow$ most influential
- **Cutting speed (A):** medium effect (range ≈ 1.034 dB)
- **Depth (C):** smallest effect (range ≈ 0.818 dB)

Approximate proportion of total effect (by range): B $\approx 57\%$; A $\approx 24\%$; C $\approx 19\%$.
 $\rightarrow$ **Feed has the largest influence on Ra** in this example, then speed, then depth.

(This matches intuition: increasing feed tends to increase roughness markedly.)

8) How Taguchi helps with robustness (interpretation)

- Taguchi's S/N uses both mean and variance in one metric: **higher S/N = lower mean & lower sensitivity to noise.**

- By maximizing S/N rather than only minimizing mean Ra, the chosen levels (A3B1C1) are expected to produce **lower Ra on average and more consistent Ra under uncontrollable noise** (tool wear, workpiece variation, environmental changes).
- The orthogonal array lets you estimate main effects with a small number of runs (efficiency) and identify the most influential factors (so you know where to focus process control).

9) Practical next steps (what Kothari / Paneerselvam style recommends)

1. **Confirmatory experiment:** run 3–5 confirmation runs at the recommended optimal settings A3B1C1 (200, 0.10, 0.5) and record Ra. Compare the measured mean & S/N to predicted values. If close, the optimization is validated.
2. If confirmation shows discrepancy, check interactions (L9 assumes main effects dominant). If interactions suspected, run a larger orthogonal array or targeted two-factor interaction experiments.
3. Implement control plan: set process default to A3B1C1 and monitor key quality metrics (Ra mean and standard deviation).

10) Summary (concise)

- Used L9 OA and the **smaller-the-better S/N** formula.
- From the simulated experimental data, **optimal settings = Cutting speed = 200 m/min; Feed = 0.10 mm/rev; Depth = 0.5 mm (A3B1C1).**
- Predicted mean Ra at optimum $\approx 0.584 \mu\text{m}$, a **~24.8%** improvement over the overall mean ($0.776 \mu\text{m}$).
- **Feed** is the most influential factor on Ra in this dataset; then cutting speed; then depth.
- Recommendation: perform a small set of confirmation runs at A3B1C1, then adopt settings if confirmed.

Q2.A fabrication company aims to improve the tensile strength of a welded joint in mild steel

components. The engineers suspect that parameters such as welding current, voltage, and

welding speed significantly affect the weld quality.

Formulate how the Taguchi method can be systematically applied to optimize the welding

process. Select an appropriate orthogonal array (e.g., L9 or L27) considering three factors at

three levels, define control factors and levels, and explain how to compute and interpret the

Signal-to-Noise (S/N) ratio for the “larger-the-better” case.

Critically evaluate how the main effect plots and ANOVA results help determine the most

influential parameters and validate the robustness of the welding process through confirmation

experiments.

Use a hypothetical dataset to illustrate your findings.

1. Problem restatement (Taguchi view)

Goal: **maximize tensile strength** of a welded joint in mild steel.
Suspected control factors (3 factors $\times$ 3 levels):

- **A — Welding current (levels A1, A2, A3)**
- **B — Welding voltage (levels B1, B2, B3)**
- **C — Welding speed (levels C1, C2, C3)**

(Units/actual numeric factor values can be filled from your process; here I use only level indices.)

Because there are 3 factors each at 3 levels, an **L9 (3^3)** orthogonal array is appropriate for an initial, efficient study.

2. L9 orthogonal array (design layout)

Standard L9 (level numbers shown):

Run A B C

1	1	1	1
2	1	2	2
3	1	3	3
4	2	1	2
5	2	2	3
6	2	3	1
7	3	1	3
8	3	2	1
9	3	3	2

Run each row with **3 replicates** (to estimate variability and compute S/N).

3. Taguchi S/N formula — “Larger-the-Better”

For a larger-is-better quality characteristic (tensile strength y) Taguchi defines the S/N ratio as:

$$S/N_{LB} = -10 \log_{10} \left(\frac{1}{n} \sum_{i=1}^n \frac{1}{y_i^2} \right)$$

where y_i are the n replicate tensile strength measurements for that run.
Higher S/N $\Rightarrow$ better (higher mean strength and less sensitivity to noise).

4. Hypothetical dataset (3 replicates per run)

I created a realistic hypothetical dataset (3 replicates/run). All numbers below are in MPa (tensile strength):

Run	A	B	C	Replicate tensile strengths (MPa)	Mean tensile (MPa)	S/N (dB)
1	1	1	1	415.873, 409.165, 409.415	411.4843	52.2863
2	1	2	2	394.781, 400.596, 391.095	395.4907	51.9415
3	1	3	3	391.234, 383.716, 386.957	387.3023	51.7602
4	2	1	2	411.252, 416.386, 405.820	411.1527	52.2786
5	2	2	3	399.033, 398.848, 403.401	400.4273	52.0501
6	2	3	1	394.700, 397.483, 395.366	395.8497	51.9505
7	3	1	3	417.127, 418.748, 413.698	416.5243	52.3925
8	3	2	1	418.434, 417.705, 416.507	417.5487	52.4141
9	3	3	2	404.703, 399.949, 401.631	402.0943	52.0862

- Grand mean tensile (over all 27 measurements) = **404.2083 MPa**
- Grand mean S/N = **52.1289 dB**

(You would replace the above hypothetical replicate values with your measured data — then follow the same calculations.)

5. Calculate factor level means (S/N & mean tensile)

Compute average S/N (and average tensile mean) at each level of each factor.

S/N means (dB) by factor-level

- A (current):
 - A1 : **51.9960 dB**

- A2 : **52.0931 dB**
- A3 : **52.2976 dB**
- B (voltage):
 - B1 : **52.3191 dB**
 - B2 : **52.1352 dB**
 - B3 : **51.9323 dB**
- C (speed):
 - C1 : **52.2170 dB**
 - C2 : **52.1021 dB**
 - C3 : **52.0676 dB**

Mean tensile (MPa) by factor-level

- A: A1 = **398.0924**, A2 = **402.4766**, A3 = **412.0558**
- B: B1 = **413.0538**, B2 = **404.4889**, B3 = **395.0821**
- C: C1 = **408.2942**, C2 = **402.9126**, C3 = **401.4180**

6. Choosing the optimal setting (Taguchi rule)

Taguchi selects the level with the **largest S/N** for each factor (since LB case). From S/N means above:

- Best **A level** = **A3** (highest S/N = 52.2976 dB)
- Best **B level** = **B1** (highest S/N = 52.3191 dB)
- Best **C level** = **C1** (highest S/N = 52.2170 dB)

So the recommended optimal combination is **A3 B1 C1**.

(Interpretation with mean tensile: this combination also coincides with high mean tensile values, consistent with S/N selection.)

7. Predicted performance at optimal setting (additive Taguchi prediction)

Taguchi prediction of the mean tensile at optimal setting uses the grand mean plus level effects:

Predicted mean tensile at A3B1C1:

$$\hat{y}_{A3B1C1} = \bar{y}_{grand} + (\bar{y}_{A3} - \bar{y}_{grand}) + (\bar{y}_{B1} - \bar{y}_{grand}) + (\bar{y}_{C1} - \bar{y}_{grand})$$

Using the numbers above:

- Grand mean tensile = **404.2083 MPa**
- $\bar{y}_{A3} = 412.0558$, $\bar{y}_{B1} = 413.0538$, $\bar{y}_{C1} = 408.2942$

Predicted tensile $\approx$ **424.9873 MPa**

Predicted S/N at A3B1C1 (same additive idea on S/N scale) $\approx$ **52.5760 dB**.

You would perform **confirmation runs** at A3B1C1 (3–5 replicates) to check if the observed mean and S/N are close to predicted values.

8. ANOVA on S/N (to quantify influence)

Perform ANOVA using S/N values (this is common in Taguchi analysis — analyze S/N to decide significance and contribution).

Summary from the hypothetical data (S/N as response):

- Sum of squares (SS) for factors:
 - $SS(A) = 0.14223$
 - $SS(B) = 0.22465$
 - $SS(C) = 0.03670$
- $SS(\text{total}) = 0.40896$ (sum of squared deviations of run S/N from grand S/N)
- Error SS = $SS(\text{total}) - (SS(A) + SS(B) + SS(C)) = 0.00538$

Degrees of freedom: for each factor $df = 2$ (3 levels $\rightarrow df=3-1$), total $df = 9-1 = 8$, error $df = 8 - (2+2+2) = 2$.

Mean squares and F ($MS_{\text{factor}} / MS_{\text{error}}$):

- $MS(A) = 0.07112 \rightarrow F(A) \approx 26.44$
- $MS(B) = 0.11232 \rightarrow F(B) \approx 41.76$
- $MS(C) = 0.01835 \rightarrow F(C) \approx 6.82$
- $MS(\text{error}) = SS_{\text{error}} / df_{\text{error}} = 0.00268999$

Percentage contribution (from SS):

- A: **34.78%**
- B: **54.93%**
- C: **8.97%**

Interpretation: voltage (B) contributes the largest portion of variation in S/N ($\approx 55\%$), followed by current (A, $\approx 35\%$), and speed (C, $\approx 9\%$). So **B is most influential**, then A, then C.

Note about statistical significance: with very small error df (2), F values can be sensitive. For formal hypothesis tests you can compare the F statistics with critical values or compute p-values. In practice, if error df is small you may want more replicates or a follow-up experiment to get more reliable error variance estimates before drawing strong significance conclusions.

9. Main-effects plots & how they help

- **Main-effects plots of S/N** show how S/N changes across the three levels for each factor. The steepness indicates effect size; a flat line indicates little effect.
- From our level means: B shows the steepest change ($B1 \gg B3$), indicating that **voltage has a big effect** on tensile robustness; A also shows substantial change ($A1 < A2 < A3$) so current matters; C is flatter (less effect).
- Main-effects plots + ANOVA together answer:
 - Which factors to control tightly (high contribution / steep slope = control key factor).
 - Which factors can be coarsely controlled or left at convenient levels (low contribution).

10. Confirmation experiment (validation)

Recommended confirmatory procedure (Kothari / Paneerselvam style):

1. Set machine to the predicted optimal levels **A3B1C1**.
2. Run **3–5 confirmation replicates** (ideally with different days / operators / material lots to test robustness).
3. Record tensile values, compute mean and S/N.
4. Compare observed mean and S/N with predicted values (predicted tensile ≈ 424.99 MPa, predicted S/N ≈ 52.576 dB).
 - If observed $\approx$ predicted (within experimental uncertainty), accept change and implement.
 - If discrepancy is large, investigate interactions or uncontrolled noise sources; you may need a larger array (e.g., L27 with interaction resolution) or targeted two-factor experiments.

Why confirm? Taguchi's additive prediction assumes negligible interactions and good experimental control — confirmation checks that assumption in practice.

11. Critical notes & best practice

- **S/N vs response ANOVA:** Taguchi recommends analyzing S/N for robustness; analyzing raw responses is also informative — they answer slightly different questions (robustness vs absolute performance). I showed S/N ANOVA above; you can also run ANOVA on the raw tensile means to get effect sizes in MPa units.
- **Interactions:** L9 estimates main effects only (confounded interactions). If you suspect important interactions between factors (e.g., current $\times$ speed), use a larger OA (L27) or a factorial follow-up to resolve interactions.
- **Error df caution:** With L9 and only 3 replicates per run, error df is small (here $df_{\text{error}}=2$). That limits the precision of MS_{error} and F tests. If you need high confidence in statistical significance, add replicates or run a bigger experiment.

12. Short numeric summary (hypothetical study)

- **Optimal (Taguchi): A3 B1 C1**
- **Predicted tensile at optimal: ≈ 425.0 MPa** (from additive prediction)
- **Predicted S/N at optimal: ≈ 52.58 dB**
- **Most influential factor: B (voltage) $\approx 55\%$ contribution; then A (current) $\approx 35\%$; C (speed) $\approx 9\%$.**
- **Recommended next step:** Run 3–5 confirmation trials at A3B1C1; if confirmed, implement and monitor. If not, run additional experiments (L27 or interaction study).

Q6. The lifetime of smartphone batteries produced by a company follows a normal distribution

with a mean $\mu = 400$ hours and standard deviation $\sigma = 40$ hours.

Find the percentage of batteries expected to last between 360 and 440 hours.

Given:

- **Mean (μ) = 400 hours**
- **Standard deviation (σ) = 40 hours**
- **We need $P(360 < X < 440)$**

Step 1: Convert to standard normal variable (Z-score)

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Compute for both limits:

Lower limit (X = 360):

$$Z_1 = \frac{360 - 400}{40} = \frac{-40}{40} = -1$$

Upper limit (X = 440):

$$Z_2 = \frac{440 - 400}{40} = \frac{40}{40} = +1$$

So we need $P(-1 < Z < 1)$.

Step 2: Use standard normal table

From Z-table:

$$P(Z < 1) = 0.8413$$

$$P(Z < -1) = 0.1587$$

Step 3: Find probability between the two Z-values

$$\begin{aligned} P(-1 < Z < 1) &= P(Z < 1) - P(Z < -1) \\ &= 0.8413 - 0.1587 = 0.6826 \end{aligned}$$

Step 4: Convert to percentage

$$0.6826 \times 100 = 68.26\%$$

Final Answer:

68.26% of batteries are expected to last between 360 and 440 hours.

Interpretation

For a normal distribution, approximately **68.26% of observations lie within $\pm 1\sigma$** from the mean.

Here, since 360 and 440 are exactly one standard deviation below and above the mean (400 ± 40), the expected percentage is **68.26%** — consistent with the empirical 68–95–99.7 rule.

Q7. The diameters of precision-made machine parts follow a normal distribution with mean $\mu =$

5.00 cm and standard deviation $\sigma = 0.02$ cm.

(a) What percentage of parts have diameters between 4.96 cm and 5.04 cm?

(b) What proportion of parts will be rejected if the acceptable tolerance limits are 4.97 cm –

5.03 cm?

Given:

- **Mean (μ) = 5.00 cm**
- **Standard deviation (σ) = 0.02 cm**
- **The variable X = diameter of parts follows a normal distribution.**

(a) Percentage of parts with diameters between 4.96 cm and 5.04 cm

Step 1: Convert limits to Z-scores

$$Z = \frac{X - \mu}{\sigma}$$

For X = 4.96:

$$Z_1 = \frac{4.96 - 5.00}{0.02} = \frac{-0.04}{0.02} = -2$$

For X = 5.04:

$$Z_2 = \frac{5.04 - 5.00}{0.02} = \frac{+0.04}{0.02} = +2$$

So we need $P(-2 < Z < 2)$.

Step 2: Use standard normal table

From Z-table:

$$\begin{aligned} P(Z < 2) &= 0.9772 \\ P(Z < -2) &= 0.0228 \end{aligned}$$

Step 3: Compute probability between the two limits

$$\begin{aligned} P(-2 < Z < 2) &= P(Z < 2) - P(Z < -2) \\ &= 0.9772 - 0.0228 = 0.9544 \end{aligned}$$

Step 4: Convert to percentage

$$0.9544 \times 100 = 95.44\%$$

Answer (a):

95.44% of parts have diameters between 4.96 cm and 5.04 cm.

(b) Proportion of parts rejected if tolerance limits are 4.97 cm – 5.03 cm

That means parts **outside this range** will be rejected.

So we need:

$$P(X < 4.97) + P(X > 5.03)$$

Step 1: Convert limits to Z-scores

For X = 4.97:

$$Z_1 = \frac{4.97 - 5.00}{0.02} = -1.5$$

For X = 5.03:

$$Z_2 = \frac{5.03 - 5.00}{0.02} = +1.5$$

Step 2: Find cumulative probabilities

From Z-table:

$$\begin{aligned} P(Z < 1.5) &= 0.9332 \\ P(Z < -1.5) &= 0.0668 \end{aligned}$$

Step 3: Compute proportion within tolerance

$$P(-1.5 < Z < 1.5) = 0.9332 - 0.0668 = 0.8664$$

So proportion accepted = 0.8664

Proportion rejected = $1 - 0.8664 = 0.1336$

Step 4: Convert to percentage

$$0.1336 \times 100 = 13.36\%$$

Answer (b):

13.36% of parts will be rejected.

Final Summary (Kothari / Paneerselvam presentation)

	Part Range (cm)	Z-Range	Probability	Result
(a)	4.96 – 5.04	–2 to +2	0.9544	95.44% within this range
(b)	4.97 – 5.03	–1.5 to +1.5	0.8664	13.36% rejected (outside limits)

Interpretation:

- About **95%** of parts fall within ± 0.04 cm of the mean (4.96–5.04 cm).
- If the company's tolerance is tightened to ± 0.03 cm (4.97–5.03 cm), then **13.36%** of parts would be **out of tolerance** and therefore **rejected**.

Q3. A researcher conducted an experiment to compare the effects of four irrigation methods (Drip, Sprinkler, Flood, and Furrow) on the growth rate of tomato plants. To minimize soil and environmental variation, a Randomized Block Design (RBD) with four blocks was used. The following ANOVA summary table was obtained from the data analysis:

Source	SS	df	MS	F
Treatment s	96.00	3	32.00	10.67
Blocks	24.00	3	8.00	2.67
Error	27.00	9	3.00	—
Total	147.00	15	—	—

Treatment means (cm/week growth): Drip = 18.5 Sprinkler = 21.0 Flood = 19.2 Furrow = 22.4

Questions:

- Based on the given F-value, interpret whether the irrigation methods differ significantly at the 5% significance level.
- Using the treatment means, determine which irrigation method produced the highest growth rate and justify your conclusion.
- Explain why the Randomized Block Design was suitable for this study, considering the potential field variability.

Given

- Treatments = 4 (Drip, Sprinkler, Flood, Furrow)
- Blocks = 4 (RBD) → each treatment appears in each block, so replications per treatment $r = 4$.
- ANOVA table (from problem):
 - SSTreat = 96.00, df_t=3, MS_t=32.00, $F_{obs} = 10.67$
 - SSBlocks = 24.00, df_b=3, MS_b=8.00, $F_{blocks} = 2.67$
 - SSerror = 27.00, df_e=9, MS_e=3.00 (so MSE = 3.00)
- Treatment means (cm/week): Drip = 18.5, Sprinkler = 21.0, Flood = 19.2, Furrow = 22.4

(a) Do irrigation methods differ at 5% significance?

We test H_0 : all treatment means equal vs H_a : at least one differs.

- From ANOVA: $F_{obs} = 10.67$ with df (3,9).
- The 5% critical value $F_{0.05,3,9} \approx 3.863$. Since $10.67 > 3.863$, reject H_0 .
- The p-value (for completeness) ≈ 0.00255 , which is < 0.05 .

Conclusion: There is a statistically significant difference among the four irrigation methods at the 5% level (follow Kothari / Paneerselvam: ANOVA shows treatment effect significant).

(b) Which method produced the highest growth and is it significantly higher?

- Highest sample mean = **Furrow = 22.4 cm/week** (so Furrow gave the largest observed growth).

To decide whether Furrow is *significantly* larger than other treatments, use the **LSD (Least Significant Difference)** method (standard in Kothari / Paneerselvam for pairwise comparisons after a significant ANOVA).

LSD formula (for RBD, equal replications r):

$$LSD = t_{\alpha/2, df_e} \times \sqrt{\frac{2 \text{ MSE}}{r}}$$

- $df_e = 9, \alpha = 0.05 \Rightarrow t_{0.025,9} \approx 2.262$.
- $\text{MSE} = 3.00, r = 4$.

Compute:

$$\text{LSD} \approx 2.262 \times \sqrt{\frac{2 \times 3}{4}} \approx 2.77 \text{ (cm/week)}$$

Now pairwise differences (absolute):

- Furrow – Drip = $22.4 - 18.5 = 3.9 \rightarrow > \text{LSD (2.77)} \rightarrow$ significant.
- Furrow – Sprinkler = $22.4 - 21.0 = 1.4 \rightarrow < \text{LSD} \rightarrow$ not significant.
- Furrow – Flood = $22.4 - 19.2 = 3.2 \rightarrow > \text{LSD} \rightarrow$ significant.

Other pairs (for completeness):

- Sprinkler – Drip = $21.0 - 18.5 = 2.5 \rightarrow < \text{LSD} \rightarrow$ not significant.
- Flood – Drip = $19.2 - 18.5 = 0.7 \rightarrow$ not significant.
- Sprinkler – Flood = $21.0 - 19.2 = 1.8 \rightarrow$ not significant.

Conclusion: Furrow produced the highest mean growth (22.4 cm/week). Using the LSD procedure ($\alpha=0.05$), Furrow is significantly higher than Drip and Flood, but not significantly higher than Sprinkler at the 5% level. So Furrow is best numerically, and significantly better than two of the three alternatives.

(c) Why was the Randomized Block Design (RBD) suitable here?

Kothari and Paneerselvam recommend blocking when experimental units have an identifiable nuisance source of variability (soil fertility, moisture gradient, shading, micro-climate, etc.). RBD is suitable because:

1. **Controls known nuisance variation:** Blocks group experimental units that are more homogeneous internally (e.g., similar soil or micro-environment). That removes between-block variability from the treatment comparison.
2. **Increases precision:** By accounting for block-to-block differences, the error (residual) variance is reduced (MSE smaller), which increases the power of the F-test to detect treatment effects.
3. **Practical & simple:** With four blocks here, each treatment appears once per block (balanced), allowing straightforward ANOVA partitioning (treatments, blocks, error).
4. **Assumptions (as in textbooks):** RBD assumes additivity of treatment and block effects, independent normal errors, and homogeneity of error variance — these should be checked/considered in interpretation.

So, because field variability (soil and environmental variation) was likely and identifiable, RBD was an appropriate design to control those nuisance effects and improve the sensitivity of the experiment.

Q4. A researcher investigates whether different study schedules affect retention in an introductory statistics course. After a common two-week module, 18 students are randomly assigned to three one-week schedules:

- **Group A – Daily Short Sessions: 30 minutes every day**
- **Group B – Alternate-Day Longer Sessions: 60 minutes every other day**
- **Group C – Single Weekly Cram: One 180-minute session at week's end**

At the end of the week, all students take the same 30-point quiz. Higher scores = better retention. Scores:

- **Group A: 22, 25, 24, 26, 23, 27**
- **Group B: 19, 20, 21, 22, 18, 20**
- **Group C: 15, 17, 16, 14, 18, 15** Using a one-way ANOVA, determine whether the groups differ significantly in quiz performance.

State clearly:

- **The null and alternative hypotheses**
- **The ANOVA table (SSA, SSW, SST, df, MS, F)**
- **The decision (Reject / Do not reject) at $\alpha = 0.05$**
- **A brief interpretation for learning/retention (and which schedule appears best)**

Groups (each $n = 6$):

- Group A (Daily short): 22,25,24,26,23,27. $\sum_A = 147$, $\bar{X}_A = 24.5$
- Group B (Alternate longer): 19,20,21,22,18,20. $\sum_B = 120$, $\bar{X}_B = 20.0$
- Group C (Weekly cram): 15,17,16,14,18,15. $\sum_C = 95$, $\bar{X}_C = 15.8333$

Total $N = 18$. Grand mean

$$\bar{X}_{..} = \frac{147 + 120 + 95}{18} = 20.111111$$

(1) Hypotheses (one-way ANOVA)

- $H_0: \mu_A = \mu_B = \mu_C$ (all population means equal).
- H_a : at least one group mean differs.

(2) Sum of squares (Kothari / Paneerselvam style)

Between (treatment) sum of squares (SSA):

$$\begin{aligned}
 SSA &= \sum_{i=1}^3 n(\bar{X}_i - \bar{X}_{..})^2 \\
 &= 6[(24.5 - 20.1111)^2 + (20.0 - 20.1111)^2 + (15.8333 - 20.1111)^2] \\
 &\approx 225.4444
 \end{aligned}$$

Within (error) sum of squares (SSE) — sum of squared deviations inside groups:

$$SSE = \sum (x_{ij} - \bar{X}_i)^2 \approx 38.3333$$

Total sum of squares:

$$SST = SSA + SSE \approx 225.4444 + 38.3333 = 263.7777$$

(3) ANOVA table

Source	SS	df	MS	F
Treatment (A,B,C)	225.4444	2	$MS_T = 225.4444/2 = 112.7222$	$F = \frac{112.7222}{2.5556} = 44.11$
Error (Within)	38.3333	15	$MS_E = 38.3333/15 = 2.5556$	
Total	263.7777	17	—	

(Values rounded in discussion; MS and F shown with 4 significant digits.)

Where $MS_E = 2.5556$ and the observed $F_{\text{obs}} \approx 44.11$.

(4) Decision at $\alpha = 0.05$

Critical value $F_{0.05,2,15} \approx 3.68$.

Since $F_{\text{obs}} = 44.11 \gg 3.68$, we **reject** H_0 .

The corresponding p-value is extremely small ($\approx 5.2 \times 10^{-7}$), so the result is highly significant.

(5) Which schedule appears best? (pairwise check — LSD)

Sample means: $\bar{X}_A = 24.5$, $\bar{X}_B = 20.0$, $\bar{X}_C = 15.8333$. Group A has the highest mean.

Using Kothari / Paneerselvam's usual LSD after a significant ANOVA:

$$LSD = t_{0.025, df=15} \sqrt{\frac{2 MS_E}{n}}.$$

With $t_{0.025,15} \approx 2.1314$, $MSE = 2.5556$, $n = 6$,

$$LSD \approx 1.97$$

Pairwise absolute differences:

- $A - B = 24.5 - 20.0 = 4.5 > 1.97 \rightarrow \text{significant}$
- $A - C = 24.5 - 15.8333 = 8.6667 > 1.97 \rightarrow \text{significant}$
- $B - C = 20.0 - 15.8333 = 4.1667 > 1.97 \rightarrow \text{significant}$

All three pairwise differences are larger than the LSD; thus every pair differs significantly at $\alpha = 0.05$.

(6) Interpretation (learning / retention)

- The one-way ANOVA shows a highly significant difference in quiz scores among the three study schedules.
- Group A (Daily short 30-min sessions) produced the highest mean quiz score (24.5 out of 30) and is **significantly better** than both Group B and Group C.
- Group B (alternate-day 60-min) is intermediate (mean 20.0) and significantly better than Group C.
- Group C (single weekly cram) performed worst (mean ≈ 15.83).

Practical conclusion: Spacing study into daily short sessions (Group A) produced the best retention for this sample and differs significantly from the other schedules. This supports the spaced practice / distributed learning idea: frequent shorter sessions improve retention relative to cramming.

Q5.A cognitive scientist tests whether practice type influences simple reaction time after a shared baseline training. 20 participants are randomly assigned for one week to one of four regimens (n's slightly unequal):

-
-
-
-

Group 1 – Visual-Only Drills (n=5): 50 trials/day of visual cues

Group 2 – Auditory-Only Drills (n=5): 50 trials/day of auditory cues

Group 3 – Multimodal Drills (n=6): 25 visual + 25 auditory trials/day

Group 4 – No Practice (n=4): No additional drills

After the week, each participant completes a 40-trial test with mixed cues. Record each person's mean reaction time (ms). Lower times = better performance.

Data (ms):

-

G1: 285, 300, 295, 310, 290

-

G2: 305, 320, 315, 325, 318

-

G3: 270, 265, 275, 280, 272, 268

-

G4: 330, 340, 335, 345

Using a one-way ANOVA, test whether mean reaction times differ among practice types. State clearly:

-

The null and alternative hypotheses

-

The ANOVA table (SSA, SSW, SST, df, MS, F) — handle unequal group sizes correctly

-

The decision at $\alpha = 0.05$

-

A brief interpretation about practice effectiveness and which regimen seems most beneficial (smaller = better)

Here's the **one-way ANOVA solution** (step-by-step, following Kothari & Panerselvam's structure):

Given data

Group	Practice Type	Reaction times (ms)	n	Mean
G1	Visual-only	285, 300, 295, 310, 290	5	296.0
G2	Auditory-only	305, 320, 315, 325, 318	5	316.6
G3	Multimodal	270, 265, 275, 280, 272, 268	6	271.7
G4	No practice	330, 340, 335, 345	4	337.5

Overall mean (grand mean) = **304.1 ms**

1. Hypotheses

- **Null hypothesis (H_0):**
($\mu_1 = \mu_2 = \mu_3 = \mu_4$)
(Mean reaction times are the same for all practice types.)
- **Alternative hypothesis (H_1):**
At least one group mean differs.

2. ANOVA Computations

Source	SS	df	MS	F	p-value
Between Groups	11807.02	3	3935.67	73.43	< 0.001
Within Groups	857.53	16	53.60		
Total	12664.55	19			

3. Decision ($\alpha = 0.05$)

From F-table:

($F_{\text{critical}}(3,16,0.05) \approx 3.24$)

Since $F = 73.43 > 3.24$,

→ **Reject H_0**

4. Interpretation

There is a **significant difference** in mean reaction times among the four practice types ($p < 0.05$).

Comparing group means (smaller = better):

Group	Mean Reaction Time (ms)	Ranking
G3 (Multimodal)	271.7	best
G1 (Visual-only)	296.0	
G2 (Auditory-only)	316.6	
G4 (No practice)	337.5	Slowest

Conclusion:

- The type of practice significantly affects reaction time.
- **Multimodal drills (both visual and auditory)** lead to the fastest responses, showing the greatest improvement in performance.
- **No-practice group** performs the worst, confirming the benefit of targeted training.

Summary (Kothari/Paneerselvam ANOVA format)

Source	SS	df	MS	F	Result
Between groups	11807.02	3	3935.67	73.43	Significant
Within groups	857.53	16	53.60		
Total	12664.55	19			

At $\alpha = 0.05$, reject $H_0 \rightarrow$ *Practice type significantly influences reaction time.*